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Reproductive Biology, Captive Breeding and Larval Development of the Threatened Deocata Pipefish *Microphis deocata* (Syngnathidae)

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Abstract—*Microphis deocata* is the only freshwater syngnathid species from the Ganges and Brahmaputra River basins in India. However, this species suffers from rampant wild collections primarily as a lucrative ornamental freshwater fish of northeast India. Given this, the present study is undertaken to manage the threat primarily posed by aquarium fish trade through the initiation of culture and breeding of the species in aquaria for the first time. The paper also highlights important notes on reproductive biology and provides a baseline datum on the spawning and larval development of *M. deocata*. A female-biased sex ratio of 1.00 : 1.24 is observed based on $n = 858$ individuals caught-and-released on site from the Manas River (Brahmaputra basin), Assam, India. Data on rearing and successful breeding trials conducted with additional 15 individuals in aquaria (six males and nine females) between June–August, 2018 and October–November, 2018 are provided. The newly-hatched larvae measures ~14 mm in total length. Transformation of larvae to juveniles commence by 75 days post hatching. After 240 days, juveniles attain a maximum length of 11.9 ± 0.1 cm weighing 0.4–0.5 g, and the mean survival rate is 56.8 ± 8.7%.

Keywords: *Microphis deocata*, Syngnathidae, biology, development, breeding, conservation, India

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INTRODUCTION

Syngnathids, which include pipefish, seahorse and seadragon, comprise 57 genera and approximately 298 species (Nelson et al., 2016). These fishes exhibit a reversal of sex roles during the mating and brooding stages where the female deposits their eggs in the male's brood pouch (Breder and Rosen, 1966; Dawson, 1985). Accordingly, syngnathids either exhibit "Gastrophori" (a brood pouch within abdomen) or "Urophori" (a brood pouch along the tail region) (Saikia et al., 2020). The female carries no further parental role after mating (Berglund et al., 1986). Mating patterns within this group exhibit significant variation: certain pipefish species demonstrate either simultaneous or sequential polygamy (Berglund et al., 1988; Rosenqvist, 1993), while other pipefish species and all investigated seahorses display monogamous behavior (Gronell, 1984; Vincent et al., 1992; Vincent and Sadler, 1995; Masonjones and Lewis, 1996; Jones et al., 1999). Unlike other fish groups, members of the family Syngnathidae have an unusual reproductive strategy (Lyons and Dunne, 2005). Some of them will carry eggs depending upon the capacity of the pouch (Gasparini and Teixeira, 1999).

Pipefishes of the genera *Microphis* mostly occur in freshwater habitats, in addition to their seawater counterparts (Dawson, 1984). *Microphis deocata* is a freshwater species native to the Ganga–Brahmaputra drainage (Froese and Pauly, 2024). Despite being the only freshwater syngnathid from the sub-Himalayan range, there are no records of the current population trend, status, reproductive biology or egg development of *M. deocata* (Froese and Pauly, 2024). The species is endemic to northeast India and categorized as Near threatened by the IUCN Red list (IUCN, 2024). This species has a huge market demand in the ornamental fish industry because of its unique body form and behaviour. Consequently, overexploitation has caused a significant loss of the wild stock. Therefore, protection of this threatened freshwater species of northeast India is crucial as it can also conserve the habitat of sympatric fish species. Therefore, the present study was undertaken to provide data on reproductive biology of *M. deocata*, and exploring the feasibility of captive breeding and rearing of this threatened freshwater species. The present work can be taken as baseline information to undertake long-term measures and management programs to compensate for the loss due

Fig. 1. Brooders of *Microphis deocata*: (a) female, 14 cm *TL*; (b) male, 11 cm *TL*.

to overexploitation of the species in wild, thus appraising towards a sustainable aquarium fish trade.

MATERIALS AND METHODS

Collection of Fishes

Live individuals were caught by conducting periodic sampling in the Manas River, Brahmaputra basin (26°28′37″ N, 90°45′46″ E), Assam (India), covering an annual cycle from March 2018 to April 2019. The specimens were caught using a scoop net (mesh size: 1.0–2.0 mm) during day hours from shallow stretches (~0.5 m depth) of the river covered with grass beds. The identity of the species was confirmed by following the taxonomic keys of Dawson (1984).

Sex Ratio

A strict catch-and-release policy was maintained for the study of sex-ratio as the species is categorized as Near Threatened (IUCN, 2024) and Schedule I (type IIA) as per the Wildlife Protection Act (1972). Sexual dimorphism was confirmed by observing the presence or absence of the brood pouch of males following the keys given by Dawson (1984) (Fig. 1). The resulting sex-ratio values were put through χ^2 analysis following the formula: $\chi^2 = \sum[(o - e)^2/e]$ (Sumbuloglu K. and Sumbuloglu V., 1998), where o is the observed value, e is the expected value.

Gonadosomatic Index, Fecundity and Oocyte Diameter

Other individuals (26 females and 24 males, $n = 50$) were additionally collected during the one-year period. These individuals were euthanized on site using clove oil (Davis et al., 2015), and then fixed in a 10% formalin solution. The fixed specimens were cleaned under running water before weighing (W, g) and quantifying for total length (TL, cm). Gonadosomatic index (GSI) was calculated separately for both the sexes using the formula: GSI = (weight of gonad (g)/body weight of fish (g)) × 100 (Singh and Srivastava, 1991).

Fecundity (F) was estimated by the gravimetric method (Hunter et al., 1989) as: $F = (N \times OW)/w$; where N is the number of ova in a sub-sample, OW is the total weight of ovary (g), and w is the weight of that sub-sample (g). The oocyte diameter was calculated by dissecting the belly of adult female specimens (TL = 12–14 cm) immediately after collection and fixation. For this, 20 mature eggs from each female were sampled.

Histological Study

Whole specimens of females fixed in 10% formalin were dissected to extract their ovaries. The ovaries were subsequently passed through ascending gradients of ethanol for dehydration and then cleared in xylene. Following embedding in paraffin wax, the ovaries were sliced into 3–4 μm sections and stained using Mayer's hematoxylin and eosin (Gutierrez, 1967). The

classification system outlined by Knorrn et al. (2024) was used to categorize the reproductive stages. To track the development of oocytes, measurements of ova diameter were taken from ovaries at all stages (Gurkan and Taylan, 2013).

Rearing, Management of Broodstock and Larvae

Live individuals (mean ± standard deviation, *SD*; $n = 15$) of brooder-sized (6 males ($TL = 11.3 \pm 0.3$ cm; $W = 0.7 \pm 0.1$ g) and 9 females ($TL = 14.1 \pm 0.3$ cm; $W = 0.9 \pm 0.9$ g)) were additionally caught in March 2018 and transported to the laboratory to study the feasibility of rearing in captivity. Before release, the fishes were treated with 2–3% salt solution for 2–5 min along with 0.1 ppm malachite green (Jithendran, 2014).

Among these, six pairs (1 male: 1 female) reared in a well-aerated glass aquarium of 213 L capacity (122 × 33 × 53 cm^3). The aquarium included a sandy substrate planted with *Vallisneria* sp. and *Hydrilla* sp., which simulated the natural habitat of the fish. All aquaria were filled with dechlorinated water and equipped with sponge and under-gravel filters. Thermostats (RS 300 W, 220–240 V) were fitted and maintained at 26°C. The brooders were fed 200 mL of *Artemia* nauplii (8–10 ind/mL^{-1}) or copepods/rotifers (6–8 ind/mL^{-1}) three times daily at 07:00, 13:00 and 18:00 h of Indian standard time (IST), respectively. The aquaria were illuminated with T5 tubes (24W) maintaining a 14 h L/10 h D cycle. Water parameters were regularly analyzed using a water quality analyzing probe (YSI Pro DSS, USA) and were maintained within the optimum range (conductivity 380–420 μS/cm^{-1}; pH 6.9–7.8; temperature 24–26°C; dissolved oxygen 6.3–6.9 g/ mL^{-1}) at par with the parameters of the natural habitat that were measured at the time of collection of the individuals.

Breeding trials commenced after three months of acclimatization of 15 brood-sized individuals. Selection and segregation of the brood pairs was made after meticulous observations, including their nuptial behavior, movement, and responses to feed and any other visual stimuli. Two breeding trials were conducted: *Trial 1*—June to August 2018; and *Trial 2*—October to November 2018. Under each trial, three sets of breeding pairs with a male-to-female ratio of 1 : 1 were included. All sets (or breeding pairs) were kept separately in six different aquaria (each aquarium with capacity 50 L) and other conditions same as describe above. After the female completed egg transfer to the male's pouch in a respective breeding set, the male was moved to a separate hatching aquarium (76 × 30 × 38 cm^3). The gestation period and number of juveniles released per male was recorded. Following hatching, the developmental stages were categorized as newborn larva, juvenile and adult. The newly-hatched larvae along with the male parent were kept in the same aquarium with continuous aeration, temperature at 26°C and a photoperiod of 14 h L: 10 h D. After the complete absorption of yolk, i.e., on the third day post hatching (dph), the larvae were fed with *Paramecium* (3 mL culture containing approximately 100 individuals per mL) at a rate of 2–3% of their body weight followed by microworm (5 mL culture containing about 15 individuals per mL), *Artemia* nauplii/plankton (10 mL culture/pond water containing 5–6 individuals per mL), chironomid larvae and *Tubifex* (15 mL culture containing 2–3 individuals per mL), thrice daily at 07:00, 13:00, and 18:00 h of IST, respectively. A detailed summary of the water management and feeding of larva is illustrated in Fig. 2. Larvae ($n = 3$) were randomly collected, observed, and measured for studying the development. Photographs of live specimens were taken under a stereo-zoom microscope (Zeiss Stemi 305) and with digital camera from hatching to 240 dph at intervals of 25, 50, 21 and 144 days.

RESULTS

Sex Ratio

A total of 858 adult individuals were caught, of which 382 were males (11–11.6 cm) and 476 were females (13.8–14.4 cm). *TL* were significantly different between the sexes ($p < 0.01$). The sex ratio of the species collected from the wild was significantly different from the expected value of 1 : 1 when the data were pooled for the whole year (i.e., male: female = 1.00: 1.24; $\chi^2 = 10.29$, $p < 0.01$).

Gonadosomatic Index

Upon determining the mean values of *GSI* for both adult male and female *M. deocata*, the highest peak was observed in July and August while the lowest was in February and March (Table 1). The mean *GSI* ranged from 8.3 ± 0.1 to 12.1 ± 0.8 for adult females, while it ranged from 5.3 ± 0.5 to 7.5 ± 0.1 for adult males.

Fecundity and Oocyte Diameter

The formula-based calculation of fecundity values for various lengths revealed a strong relationship ($\log F = -2.02 + 3.53 \log TL$; $R^2 = 0.9356$). The fecundity of adult females ($n = 26$; $TL = 12.1–14.9$ cm) ranged from 54 to 128 oocytes (91 ± 19). The oocyte diameter of *M. deocata* varied between 0.5–1.3 mm in the ripe stage.

Macroscopic and Microscopic Description of Ovary

The ovaries are bilaterally paired structures that undergo fusion in the terminal segment of the body axis, occupying approximately 75% of the female trunk length. In mature females (14.1 ± 0.3 cm), the ovarian dimensions were recorded to be 28 mm in length and a mass of 0.16 g, demonstrating bilateral asymmetry. Notably, 57% of the observed females exhibited discernible asymmetry, wherein the right

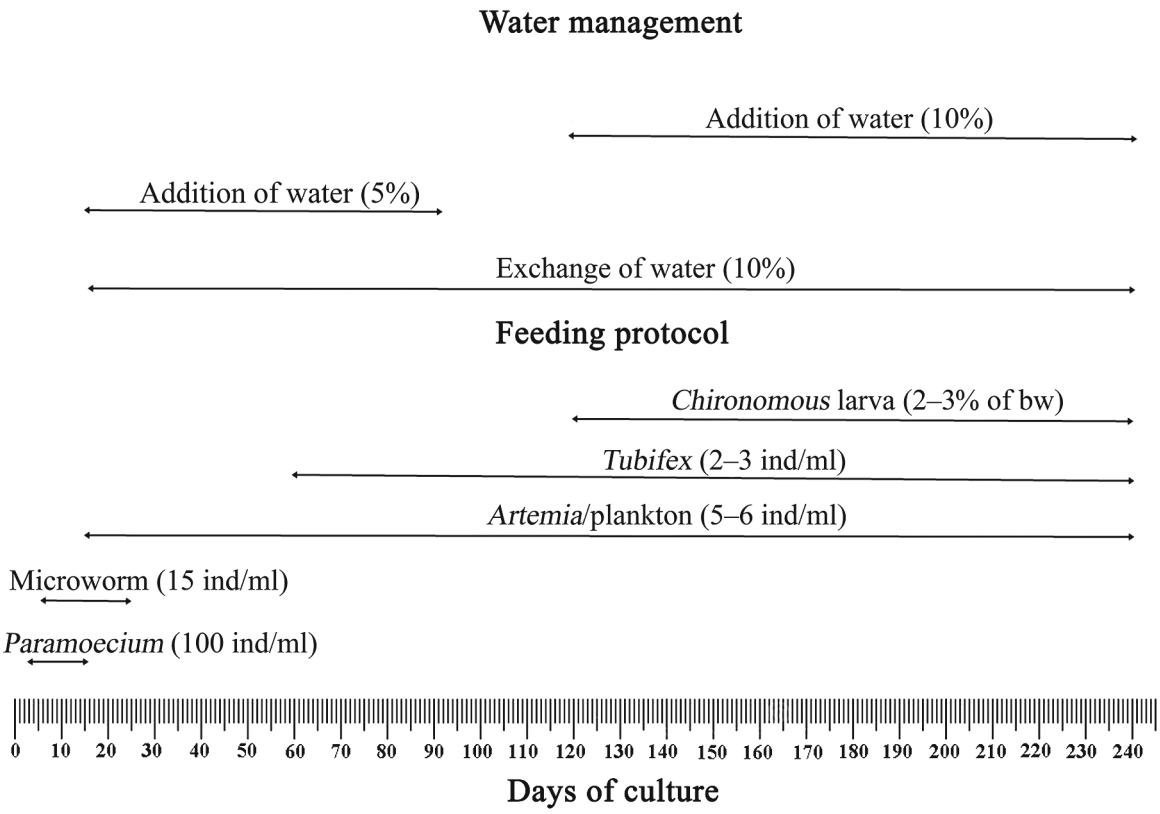

Fig. 2. Water quality management and feeding protocol of larval rearing of *Microphis deocata*.

ovary demonstrated a marginal elongation relative to the left (0.3–2.8 mm difference). A detailed classification of the ovary in different reproductive stages is shown in Table 2 and Fig. 3.

Spawning Initiation and Larval Development

After three months of captive rearing, ripe females of *M. deocata* began to show pre-spawning displays in the early morning hours by inflating the abdomen about 2 cm from their body axis. Subsequently, the males started to chase the females. This pre-spawning display lasted for 1–2 h (Fig. 4a). No postural displays or courtship were observed for the rest of the day. Following this observation, the active pairs were selected and segregated to conduct future breeding trials. Soon after transfer the next morning, both partners swam side-by-side towards the water surface and initiated

Table 1. Monthly variation (mean ± *SD*) in body weight, ovary weight, testis weight and gonadosomatic index (*GSI*) of female and male of *Microphis deocata* ($n = 2$ for both sexes in each month) collected from Manas River (March 2018 and April 2019)

Month	Female			Male		
	Body weight	Ovary weight	*GSI*	Body weight	Testis weight	*GSI*
January	0.860 ± 0.060	0.079 ± 0.010	9.25 ± 0.12	0.670 ± 0.020	0.036 ± 0.001	5.35 ± 0.09
February	0.840 ± 0.060	0.077 ± 0.005	9.17 ± 0.12	0.660 ± 0.020	0.036 ± 0.003	5.39 ± 0.2
March	0.770 ± 0.050	0.064 ± 0.004	8.30 ± 0.11	0.660 ± 0.040	0.034 ± 0.001	5.26 ± 0.52
April	0.800 ± 0.070	0.078 ± 0.006	9.79 ± 0.07	0.660 ± 0.040	0.037 ± 0.002	5.67 ± 0.07
May	0.800 ± 0.040	0.082 ± 0.004	10.16 ± 0.11	0.650 ± 0.040	0.043 ± 0.003	5.90 ± 0.12
June	0.800 ± 0.140	0.082 ± 0.017	10.28 ± 0.45	0.670 ± 0.050	0.044 ± 0.003	6.53 ± 0.08
July	1.040 ± 0.090	0.122 ± 0.012	12.11 ± 0.82	0.650 ± 0.070	0.047 ± 0.006	7.23 ± 0.19
August	0.930 ± 0.070	0.100 ± 0.010	11.37 ± 0.42	0.630 ± 0.090	0.047 ± 0.007	7.46 ± 0.06
September	0.970 ± 0.080	0.100 ± 0.009	10.77 ± 0.06	0.600 ± 0.007	0.039 ± 0.002	6.37 ± 0.35
October	0.960 ± 0.030	0.090 ± 0.004	9.76 ± 0.13	0.600 ± 0.008	0.037 ± 0.001	6.21 ± 0.14
November	0.840 ± 0.080	0.070 ± 0.006	9.23 ± 0.12	0.600 ± 0.010	0.035 ± 0.001	5.76 ± 0.09
December	0.860 ± 0.080	0.070 ± 0.007	8.80 ± 0.09	0.600 ± 0.010	0.033 ± 0.007	5.44 ± 0.04

Table 2. Classification of reproductive stages of ovary of *Microphis deocata*

Reproductive stage	Total length, cm	Macroscopic characteristic	Microscopic feature
Immature stage (I)	6.8 ± 2.4	The ovaries exhibit characteristics of being diminutive, slender, and transparent	Oocytes in the O and PvO stages are exclusively present; atresia is not seen. Oocytes were packed with each other with minimal space
Early developing (II)	8.2 ± 0.5	Ovaries occupying 1/4 of the trunk region	Both PvO and CA oocytes were present
Developing (III)	9.8 ± 1.2	Enlarged ovaries occupying 3/4 of the trunk region	PvO, CA, Vtg1, Vtg2 oocytes were seen, atresia is seen
Spawning capable (IV)	11.6 ± 2.8	Expansive ovaries occupy a significant portion of the peritoneal cavity, with prominent blood vessels, and individual oocytes are visible to the naked eye	Vtg3 oocytes were present along with PvO, CA, Vtg1 and Vtg2. Atresia is also seen
Actively spawning (V)	14.1 ± 0.3	The peritoneal cavity is completely filled by the ovaries, and the body walls generally exhibit expansion. The oocytes are delicate and susceptible to easy tearing	Oocyte hydration takes place. PvO, CA, Vtg1, Vtg2 and Vtg3 oocytes were seen

O, oogonia; PvO, pre-vitellogenic oocyte; CA, cortical alveoli; Vtg 1–Vtg 3, primary to tertiary vitellogenic oocyte; VO, vitellogenic oocyte.

mating behavior. The male frequently nudged the belly of the female with his head and opened his brood pouch by swaying his tail forward towards his belly. Following this, the female deposited her eggs (Fig. 4b). Courtship displays included intertwining among themselves like a twisted rope, however, exhibiting distinct *S*-shaped curves.

Among the six breeding sets, the "second set" was unsuccessful in *Trial 1* (i.e., June–August, 2018); whereas the "third set" was unsuccessful in *Trial 2* (i.e., from October–November, 2018). Details of both the breeding trials (inclusive of all six sets) are summarized in Table 3.

Before releasing the juveniles, the male became inactive, hiding in darker areas of the aquarium. Release of juveniles continued up to three days and each male released up to 16–18 larvae. The gestation time was observed to be 22.8 ± 1.7 days. Before hatching, the embryos stay coiled within the egg envelope (Fig. 5a). The newly-hatched larvae measured ~14 mm in *TL* with less yolk-sac and were free-swimming (Fig. 5b). The larvae usually hatch during the morning hours and the hatching percentage was 91.8 ± 3.5. However, the survival rate was very low (i.e., 56.8 ± 8.7%) in the present study.

The head of the newly-hatched larvae presents a slight bend with respect to the body axis unlike in adults, wherein the head and snout are oriented longitudinally. The newly-hatched larvae tend to swim at the bottom of the aquarium and remain hidden in vegetation, which indicates a benthopelagic strategy. At 25 dph, the larvae attained a *TL* of 22 mm with completely resorbed yolk (Fig. 5c). The snout measured about 47.1% of head length (*HL*). A total of 45 body ridges were present: 20 in the trunk and 25 in the tail region. The dorsal and caudal fins include 34 and 8 fin rays respectively. The pectoral primordium is translucent and webbed-like where fine veins emerge in a branching pattern. At 75 dph, the larvae attained a *TL* of 37 mm (Fig. 5d) and were considered juveniles. At 96 dph, it attained 45 mm *TL* (Fig. 5e) with snout cover occupying 50% *HL*. The bony ridges in the trunk and tail regions (51 in number) transformed into complete body rings. In this form, the pectoral anlage converts into a pectoral fin with 9 fin rays. At 240 dph, the larvae attained 119 mm *TL* with a snout cover of 46.1% *HL* (Fig. 5f). At this stage, 56 bony rings in the body and 26 fin rays in the pectoral fin were observed.

DISCUSSION

A female-biased sex ratio of 1.00 : 1.24 was observed in *Microphis deocata*, which is similar to as reported in *Syngnathus abaster* (1.0 : 2.7) and *S. typhle* (1.0 : 1.2) (Simal Rodríguez, et al., 2021). In *S. biaculeatus*, the sex ratio was female-biased in autumn (Takahashi et al., 2003), while it was unbiased for the rest seasons. Such female-biased populations will eventually give rise to a more competitive group for the available males resulting in lesser male crowding than females (Aronsen et al., 2013).

The range in oocyte diameter at different stages of development *M. deocata* was 0.5–1.3 mm; while, in *S. abaster*, it ranged from 1.09 to 2.06 mm (Silva et al., 2006), and, in *S. acus*, it ranged from 0.4 to 1.8 mm (Gurkan et al., 2009). The average fecundity of *M. deocata* is much lesser when compared to ranges of 100–500 in other syngnathids as per previous studies

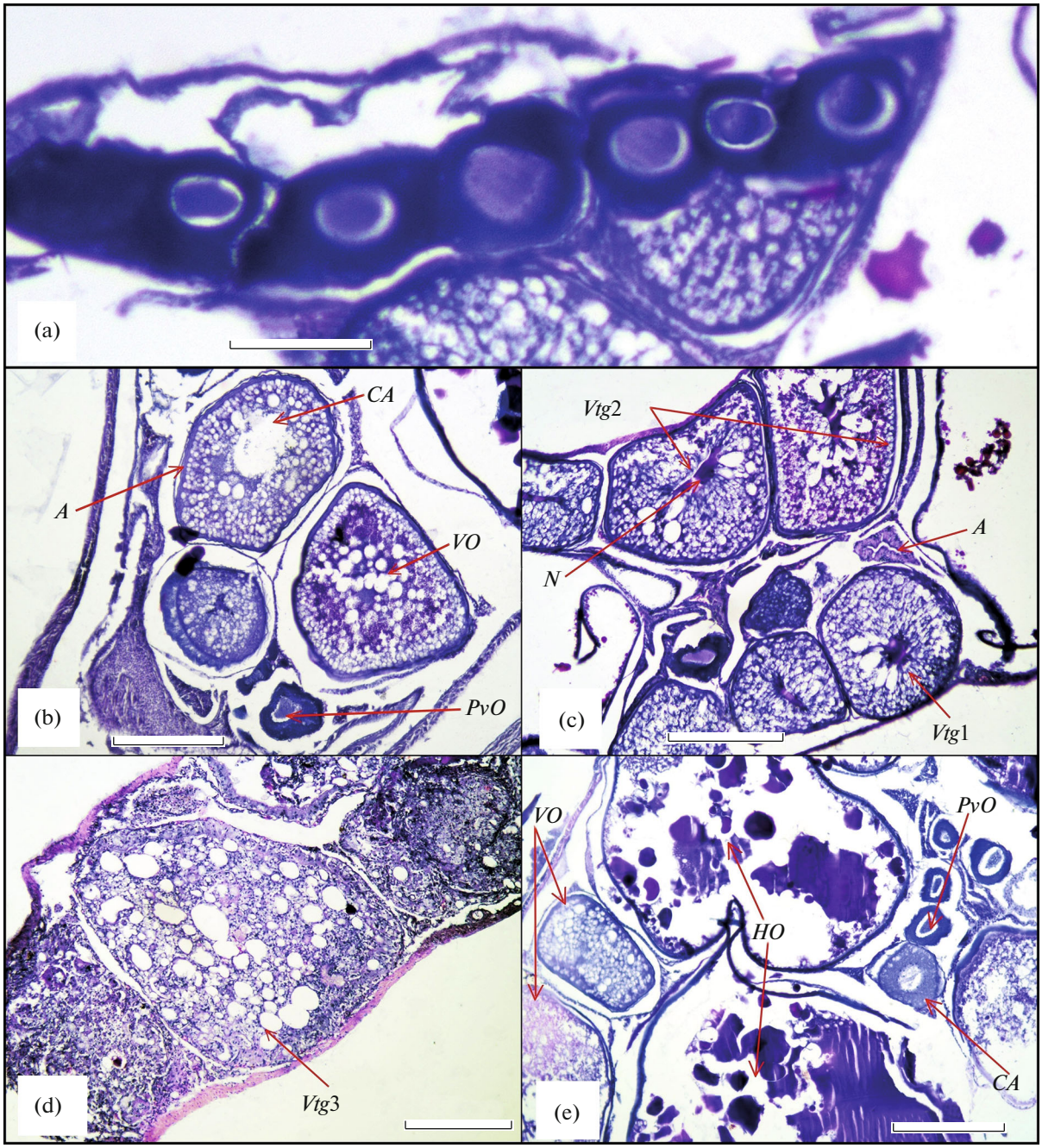

Fig. 3. Histological cross sections of ovarian development of *Microphis deocata* (reproductive stage based on Table 4): (a) immature stage; (b) early developing; (c) developing; (d) spawning capable; (e) actively spawning. *PvO*, pre-vitellogenic oocyte; *A*, atresia; *CA*, cortical alveoli; *VO*, vitellogenic oocyte; *N*, nucleus; *HO*, hydrated oocyte; *Vtg*1–*Vtg*3, primary to tertiary vitellogenic oocyte. Scale, μm: (a) 100; (b, c) 50; (d, e) 500.

(Dawson, 1985; Monteiro et al., 2005). However, each spawning event yielded approximately 20 eggs, indicative of a batch-spawning reproductive strategy.

Microphis deocata breeds throughout the year, as evidenced during collection with several male individuals to always bear embryos. But, variations in *GSI* values show an increased reproductive potential from June to October with a spawning peak in July and August. Period of reproduction is variable among different species of pipefishes across their distributional ranges. For example, the reproductive period of *S. acus* in Camalti Lagoon (Izmir Bay, Aegean Sea) varies between March and September (Gurkan et al., 2009) or varies between four to nine months in other syngnathids in Moreton Bay (Queensland, Australia) (Takahashi et al., 2003). A study suggests a lack of a

Fig. 4. Spawning behavior of female 14 cm *TL* (a) and courtship display in *Microphis deocata* (male 11 cm *TL*; female 14 cm *TL*) (b).

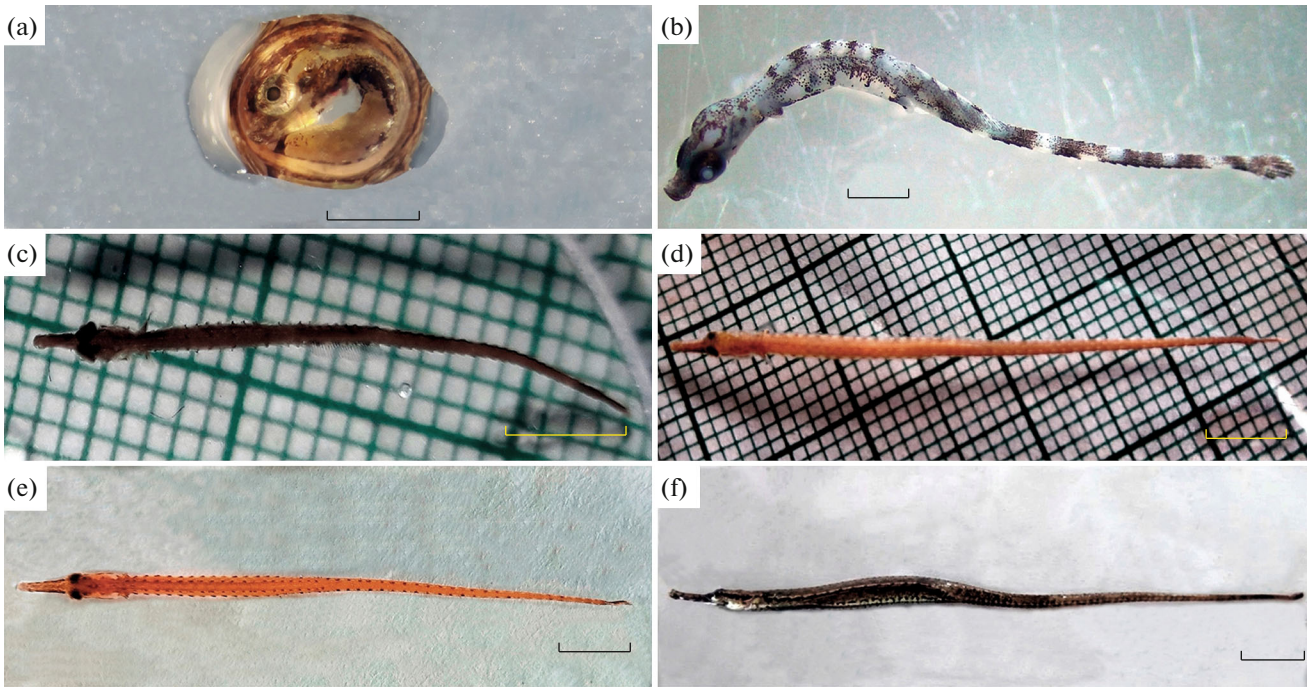

Fig. 5. Larvae and juveniles of *Microphis deocata*: coiled-larva within the chorion (a); newly-hatched larva 14 mm *TL* (b); larva: at 25 days post hatching (dph), 22 mm *TL* (c); 75 dph, 37 mm *TL* (d); 96 dph, 45 mm *TL* (e); 240 dph, 119 mm *TL* (f). Scale, mm: (a) 0.5, (b) 1.0, (c–e) 5.0, (f) 10.0.

Table 3. Summary of successful captive breeding trials of *Microphis deocata* in aquaria

Parameter	June–August, 2018						October–November, 2018						Mean ± *SD* of all trials	
	*Trial 1**						*Trial 2**							
	Set 1		Set 2		Set 3		Set 1		Set 2		Set 3			
	male	female	male	female	male	female	male	female	male	female	male	female	male	female
Total length, cm	10.9	14.3	11.5	13.7	11.0	14.0	11.5	13.9	11.1	14.3	10.8	13.7	11.3 ± 0.3	14.1 ± 0.3
Body weight, g	0.6	0.9	0.7	0.7	0.7	0.8	0.6	0.8	0.6	0.9	0.5	0.6	0.7 ± 0.1	0.9 ± 0.1
Spawning initiation time	4:30–5:40		×		4:20–5:30		5:00–6:00		5:00–6:00		×			
No. of eggs deposited in male	22		×		20		18		21		×		20.3 ± 1.7	
No. of fertilized eggs	19		×		20		17		19		×		18.8 ± 1.3	
Fertilization rate, %	86.4		×		100		94.4		90.5		×		92.8 ± 5.8	
Incubation period, days	22		×		21		25		23		×		22.8 ± 1.7	
No. of hatched larva	17		×		19		15		18		×		17.3 ± 1.7	
Hatching rate, %	89.4		×		95.0		88.2		94.7		×		91.8 ± 3.5	
No. of surviving adults	8		×		10		10		11		×		9.8 ± 1.3	
Survival rate, %	47.0		×		52.6		66.6		61.1		×		56.8 ± 8.7	

Set 1–Set 3, brooders; "×", indicates unsuccessful captive breeding; * see in the text.

Table 4. Reproductive characteristics of freshwater pipefishes of the genus *Microphis* (reproductive season and time refers to the period of day when courtship and spawning behavior were observed)

Species	Origin	Reproductive season	Time	Reference
Microphis aculeatus	Africa	November	Morning	Christie, 2022
Microphis lineatus	North America	May–November	–	Frias-Torres, 2004
Microphis lineatus	North America	Summer, Winter	–	Miranda-Marure et al., 2004
Microphis leaispis	Japan	June–December	–	Ishihara and Tachihara, 2008
Microphis deocata	India	June–August and October–November	Morning	Present study

"–", unavailability of data.

specific reproductive season in *M. brachyurus* (Miranda-Marure et al., 2004). Another study reports *M. lineatus* to have two reproductive seasons in the Brazilian population, while continual reproduction from May to November was observed in the Caribbean population (Miranda-Marure et al., 2004). Whereas, Perrone (1989) confirmed that *M. brachyurus* reproduces throughout the year. Nevertheless, reproductive season varies among closely-related species of the genus *Microphis*, which is summarized in Table 4.

Through histological analysis of the ovaries of *M. deocata*, we observed a lack of synchrony in oocyte development similar to as found in *S. abaster* and *S. typhle* (Simal Rodriguez et al., 2021). This suggests that the species is a batch spawner (Kornienko, 2001). Furthermore, females exhibit vitellogenic oocytes consistently throughout all seasons similar to as found in *M. brachyurus* (Miranda-Marure et al., 2004). Fish species that follow a semelparous reproductive strategy, characterized by a single spawning season, do not retain a reserve of previtellogenic oocytes (PvO). Instead, all PvO oocytes progress into secondary growth oocytes (Lowerre-Barbieri et al., 2011). This is in contrast to our study where primary growth oocytes were seen in every reproductive stage indicating iteroparity in reproduction.

The pre-spawning behavior of *M. deocata* (Saikia et al., 2024) was similar to that described for *Stigmatophora nigra* and *S. argus* (Rosenqvist and Berglund, 2011). However, the duration of this behavior until spawning was longer, pre-spawning displays lasted for 1–2 h in *M. deocata* and in *M. aculeatus*, it began 9 to 11 min after first light and continued for 19–26 min and lasted until 40 min (Christie, 2022). No postural displays were observed through rest of the day, which is similar to as reported in *M. aculeatus*. In other syn-

gnathids, like the sea horse *H. guttulatus*, the display extends all day long (Planas et al., 2008). Pipefishes engage in courtship displays, intertwining like a twisted rope. However, the distinctive *S*-shaped curves displayed by *M. deocata* bear similarities to those observed in *M. aculeatus* and *S. abaster* (Breder and Rosen, 1966; Silva et al., 2006; Christie, 2022).

Microphis deocata exhibits a notably low survival rate potentially attributed to the ingestion of air bubbles while preying on brine shrimp (*Artemia*), a similar phenomenon also observed in *H. abdominalis* (Woods, 2000).

Unlike other Gastrophori larvae of *M. leiaspis,* the newly-hatched larvae of *M. deocata* resemble adults in terms of morphology (Ishihara and Tachihara, 2008). Upon hatching, the larvae exhibit a preference for swimming close to the bottom of the aquarium and seek concealment within vegetation, adopting a benthopelagic strategy akin to that observed in *S. abaster*.

CONCLUSIONS

In conclusion, this study provides baseline data on the successful captive breeding and rearing of *M. deocata* in aquaria. Additionally, important notes on reproductive biology of the species have also been added. From the study, it was clear that captive breeding and rearing of the species is feasible in aquaria provided live feed is available *ad libitum* for rearing the larva. As *M. deocata* is a potentially lucrative, high-demanding freshwater ornamental fish of northeast India, the development of a farmer-friendly culture protocol will reduce its indiscriminate exploitation from the wild, besides generating livelihood for the rural poor. However, the associated problems of the species' culture, such as low survival rates, require further research and development to be addressed.

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ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The present study was approved by the Institutional Animal Ethical Committee of Gauhati University, Assam (Reference ID: IAEC/PER/2019/PP-IAEC/2018-034).

CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

REFERENCES

Aronsen, T., Mobley, K.B., Berglund, A, et al., The operational sex ratio and density influence spatial relationships between breeding pipefish, *Behav. Ecol.,* 2013, vol. 24, no. 4, pp. 888–897.
https://doi.org/10.1093/beheco/art019

Berglund, A., Rosenqvist, G., and Svensson, I., Reversed sex roles and parental energy investment in zygotes of two pipefish (Syngnathidae) species, *Mar. Ecol. Prog. Ser.,* 1986, vol. 29, pp. 209–215.
https://doi.org/10.3354/meps029209

Berglund, A., Rosenqvist, G., and Svensson, I., Multiple matings and paternal brood care in the pipefish *Syngnathus typhle, Oikos*, 1988, vol. 51, no. 2, pp. 184–188.
https://doi.org/10.2307/3565641

Breder, C.M., and Rosen, D.E., *Modes of Reproduction in Fishes*, New York: Nat. Hist. Press, 1966.

Christie, B.L., Courtship behaviour of the freshwater pipefish *Microphis aculeatus* (Syngnathidae): A case study in captive breeding, *J. Zool. Aquar. Res.*, 2022, vol. 10, pp. 47–53.
https://doi.org/10.19227/jzar.v10i1.545

Davis, D.J., Klug, J., Hankins, M., et al., Effects of clove oil as a euthanasia agent on blood collection efficiency and serum cortisol levels in *Danio rerio, J. Am. Assoc. Lab. Anim. Sci.*, 2015, vol. 54, 564–567.

Dawson, C.E., Revision of the genus *Microphis* Kaup (Pisces: Syngnathidae), *Bull. Mar. Sci.*, 1984, vol. 35, pp. 117–181.

Dawson, C.E., *Indo-Pacific Pipefishes, Red Sea to the Americas*, Ocean Springs, MS: The Gulf Coast Res. Lab., 1985.

Frias-Torres, S., Notes on aquarium brood release and feeding of the opossum pipefish, *Microphis brachyurus* lineatus, *Gulf Caribb. Res.,* 2004, vol. 16, pp. 73–75.
https://doi.org/10.18785/gcr.1601.11

Froese, R., and Pauly, D., *Fishbase, Version 02/2024*, 2024. www.fishbase.org, Gasparini, J.L. and Teixeira, R.L., Reproductive aspects of the Gulf pipefish *Syngnathus scovelli* (Teleostei: Syngnathidae), from Southeastern, Brazil, *Braz. J. Biol.,* 1999, vol. 59, no.1, pp. 87–90.
https://doi.org/10.1590/S0034-71081999000100011

Gronell, A.M., Courtship, spawning and social organization of the pipefish, *Corythoichthys intestinalis* (Pisces: Syngnathidae) with notes on two congeneric species, *Z. Tierpsychol.*, 1984, vol. 65, no. 1, pp. 1–24.

Gürkan, Ş. and Taylan, B., The first observations of oocyte diameters for the pipefish from the southern Black Sea, *J. Black Sea/Medit. Environ.*, 2013, vol. 19, no. 2, pp. 206–216.

Gurkan, S., Taskavak, E., and Hossucu, B., The reproductive biology of the Great Pipefish *Syngnathus acus* (Family: Syngnathidae) in the Aegean Sea, *North West J. Zool.,* 2009, vol. 5, no. 1, pp. 179–190.

Gutierrez, M., Coloracion histologica para ovarios de peces, crustaceos y moluscos, *Invest. Pesq.*, 1967, vol. 31, no. 2, pp. 265–271.

Hunter, J.R., Macewicz, B.J., and Kimbrell, C.A., Fecundity and other aspects of the reproduction of sablefish, *Ampoporna fimbria*, in Central California waters, *CalCOFI Report.*, 1989, vol. 30, pp. 61–72

Ishihara, T. and Tachihara, K., Reproduction and early development of a freshwater pipefish *Microphis leiaspis* in Okinawa–Jima Island, Japan, *Ichthyol. Res.,* 2008, vol. 55, pp. 349–355.
https://doi.org/10.1007/s10228-008-0049-y

IUCN, *The IUCN Red List of Threatened Species, Version 02/2024*, 2024. https://www.iucnredlist.org.

Jithendran, K.P., *Parasites and parasitic diseases in fish culture system*, *Veterinar. Parasitol.*, 2014, ch. 12, pp. 331–376. https://doi.org/krishi.icar.gov.in

Jones, A.G., Rosenqvist, G., Berglund, A., and Avise, J.C., The genetic mating system of a sex-role-reversed pipefish (*Syngnathus typhle*): A molecular inquiry, *Behav. Ecol. Sociobiol.*, 1999, vol. 46, pp. 357–365.
https://doi.org/10.1007/s002650050630

Knorrn, A.H., Wieben, K.L., Fock, H.O., and Andresen, H., Reproductive biology of the electric lanternfish *Electrona risso* (Myctophidae) and the bigscale fishes *Melamphaes polylepis* and *Scopelogadus mizolepis* (Melamphaidae), *J. Fish. Biol.*, 2024, vol. 104, no. 1, pp. 252–264.
https://doi.org/10.1111/jfb.15575

Kornienko, E.S., Reproduction and development in some genera of pipefish and seahorses of the family Syngnathidae, *Russ. J. Mar. Biol.,* 2001, vol. 27, pp. S15–S26.
https://doi.org/10.1023/A:1013894804529

Lowerre-Barbieri, S.K., Brown-Peterson, N.J., Murua, H., et al., Emerging issues and methodological advances in fisheries reproductive biology, *Mar. Coast. Fish.*, 2011, vol. 3, pp. 32–51.
https://doi.org/10.1080/19425120.2011.555725

Lyons, D.O. and Dunne, J.J., Reproductive ecology and operational sex ratio of Worm Pipefish (*Nerophis lumbriciformis)* in Irish waters. In biology and environment, *Proc. Royal Irish Acad.*, 2005, vol. 105, no. 1, pp. 9–14.
https://doi.org/10.3318/BIOE.2005.105.1.9

Masonjones, H.D. and Lewis, S.M., Courtship behavior in the dwarf seahorse, *Hippocampus zosterae, Copeia*, 1996, no. 3, pp. 634–640.
https://doi.org/10.2307/1447527

Miranda-Marure, M.E., Martinez-Perez, J.A., and Brown-Peterson, N.J., Reproductive biology of the opossum pipefish, *Microphis brachyurus lineatus*, in Tecolutla Estuary, Veracruz, Mexico, *Gulf Caribb. Res.,* 2004, vol. 16, no. 1, pp. 101–108.
https://doi.org/10.18785/gcr.1601.17

Monteiro, N.M., Almada, V.C., and Vieira, M.N., Implications of different brood pouch structures in syngnathid reproduction, *J. Mar. Biol. Assoc. UK,* 2005, vol. 85, no. 5, pp. 1235–1242.
https://doi.org/10.1017/S0025315405012373

Nelson, J.S., Grande, T.C., and Wilson, M.V., *Fishes of the World*, Hoboken: John Wiley and Sons, Inc., 2016.
https://doi.org/10.1002/9781119174844

Perrone, E.C., Distribução sazonal e épocas de reprodução do peixe- cachimbo, *Oostethus lineatus* (Pisces: Sygnathidae) em um trecho do Rio Jucú, Espírito Santo, *Revista Cultur. Univer. Fed. Espírito Santo*, 1989, vol. 41, no. 42, pp. 121–126.

Planas, M., Chamorro, A., Quintas, P., and Vilar, A., Establishment and maintenance of threatened long-snouted seahorse, *Hippocampus guttulatus*, broodstock in captivity, *Aquaculture*, 2008, vol. 283, pp. 19–28.
https://doi.org/10.1016/j.aquaculture.2008.06.023

Rosenqvist, G., Sex role reversal in a pipefish, *Mar. Freshw. Behav. Phys.*, 1993, vol. 23 no. 1–4, pp. 219–230.
https://doi.org/10.1080/10236249309378867

Rosenqvist, G. and Berglund, A., Sexual signals and mating patterns in Syngnathidae, *J. Fish Biol.,* 2011, vol. 78, no. 6, pp. 1647–1661.
https://doi.org/10.1111/j.1095-8649.2011.02972.x

Saikia, A., Kalita, M., Choudhury, H., et al., Embryonic development of the threatened freshwater pipefish *Microphis deocata* (Hamilton, 1822), *J. Appl. Ichthyol.,* 2020, vol. 36, no, 1, pp. 90–94.
https://doi.org/10.1111/jai.13968

Saikia, A., Nath, J.K., and Sarma, D., Observations on the courtship behaviour of Deocata Pipefish *Microphis deocata* (Hamilton, 1822) (Actinopterygii: Syngnathiformes: Syngnathidae) in an aquarium, *J. Threat. Taxa*, 2024, vol. 16, no. 1, pp. 24529–24534.
https://doi.org/10.11609/jott.8684.16.1.24529-24534

Silva, K., Monteiro, N.M., Vieira, M.N., and Almada, V.C., Reproductive behavior of the black-striped pipefish *Syngnathus abaster* (Pisces: Syngnathidae), *J. Fish Biol.,* 2006, vol. 69, pp. 1860–1869.
https://doi.org/10.1111/j.1095-8649.2006.01229.x

Simal Rodríguez, A., Grau, A., Castro-Fernández, J., et al., Reproductive biology of Pipefish *Syngnathus typhle* and *S. abaster* (Syngnathidae) from Western Mediterranean Sea, *J. Ichthyol.*, 2021, vol. 61, no. 4, pp. 608–615.
https://doi.org/10.1134/S0032945221040147

Singh, S.R. and Srivastava, V.K., Observations on the gonadosomatic index and fecundity of Ganga River prawn *Macrobracium binnanicum* Choprai (Tiwari), *J. Adv. Zool.,* 1991, vol. 12, no. 1, pp. 50–55.

Sumbuloglu, K. and Sumbuloglu, V., *Biostatistics*, Ankara: Hatiboglu, 1998.

Takahashi, E., Connolly, R.M., and Lee, S.Y., Growth and reproduction of double-ended pipefish, *Syngnathoides biaculeatus*, in Moreton Bay, Queensland, Australia, *Environ. Biol. Fishes.,* 2003, vol. 67, pp. 23–33.
https://doi.org/10.1023/A:1024416031274

Vincent, A.C., and Sadler, L.M., Faithful pair bonds in wild seahorses, *Hippocampus whitei. Anim. Behav.*, 1995, vol. 50, no. 6, pp. 1557–1569.

Vincent, A., Ahnesjö, I., Berglund, A., and Rosenqvist, G., Pipefishes and seahorses: Are they all sex role reversed?, *Trends Ecol. Evol.*, 1992, vol. 7, no. 7, pp. 237–241.

Woods, C.M.C., Preliminary observations on breeding and rearing the seahorse *Hippocampus abdominalis* (Teleostei: Syngnathidae) in captivity, *N. Z. J. Mar. Freshw. Res.,* 2000, vol. 34, pp. 475–485.
https://doi.org/10.1080/00288330.2000.9516950

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